

Project Report: Salary Prediction System Model

Group Members:

Leader: Abu Bakar Farooq 545

- S.M. Ahsan 503
- Abdul Reheem 549
- Abdullah 555
- Abdul Rehman 504

Executive Summary

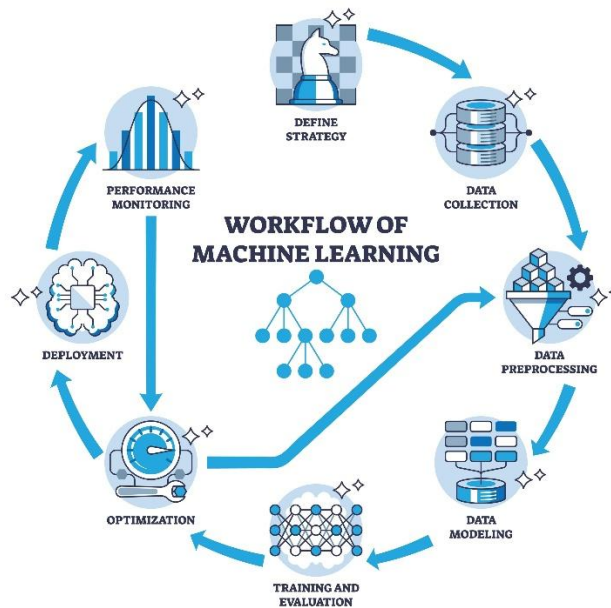
The **Intelligent Salary Prediction System** is a data-driven application designed to estimate employee compensation based on professional experience, demographic data, and educational background. By leveraging machine learning techniques, specifically **Ridge Regression**, the system identifies correlations between employee attributes and their monthly salaries. The final solution is deployed as an interactive web application using the **Streamlit** framework, allowing for real-time predictions.

1. Introduction

This project is like building a "Smart Calculator." Instead of just adding numbers, this calculator guesses how much money a person should earn. We taught a computer to look at details like a person's age, their education, and how many years they have worked. Based on these details, the computer predicts their **Monthly Salary**.

2. System Architecture

The project follows a standard Machine Learning pipeline consisting of three distinct phases: Data Ingestion, Model Training, and Deployment.



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2.1 Workflow Overview

1. **Data Preprocessing:** Raw data is cleaned and formatted using Pandas and NumPy.
2. **Model Training:** A regression algorithm learns the relationship between features (e.g., Age, Experience) and the target variable (Salary).
3. **Deployment:** The trained model is serialized and integrated into a user-friendly interface.

3.How the System Works

To make this work, we followed three simple steps:

Step 1: Gathering the Data (The "Textbook")

Before the computer can guess, it needs to study. We gave it a list of 50 employees. This list acts like a textbook. It contains:

- **Experience:** How long they have worked.
- **Age:** How old they are.
- **Education:** If they went to High School, College, or University.
- **Salary:** How much they actually get paid.

Step 2: Training the Brain (The "Code")

We used a computer program (Python) to act as a teacher.

- **Cleaning:** First, the code fixes any missing answers in the list so the computer doesn't get confused.
- **Learning:** We used a method called **Ridge Regression**. This is like a math student practicing problems until they get the answers right.
- **Saving:** Once the computer learned the patterns, we saved its "brain" into a file called `ridge_regression_model.pkl`.

Step 3: The Prediction App (The "Remote Control")

Finally, we made a website using a tool called **Streamlit**. This is the part you can see and touch.

- You type in the details (like "Age" or "Years of Work").
- You click a button.
- The app asks the saved "brain" for the answer and shows the salary on the screen.

4. Dataset Description

The model is trained on the `employee_salary_dataset.csv`, which contains historical records of 50 employees.

Key Features (Independent Variables):

- **Experience_Years:** Quantitative value representing professional tenure.
- **Age:** The biological age of the employee.
- **Education_Level:** Categorical data (High School, Bachelor, Master, PhD) utilized to weigh qualification impact.
- **Job_Level:** A scale (1–5) indicating the hierarchy or rank of the employee within the organization.

Target Variable (Dependent Variable):

- **Monthly_Salary:** The continuous variable representing the financial compensation to be predicted.

5. Technical Methodology

5.1 Data Processing

The raw data is processed using Python libraries `pandas` and `numpy`.

- **Missing Value Imputation:** To ensure data integrity, missing entries in the dataset are filled using the column mean strategy (`data.fillna(data.mean())`).

- **Feature Selection:** The dataset is split into the feature matrix (\$X\$) and the target vector (\$y\$) by dropping the "Salary" column from the input set.

5.2 Machine Learning Algorithm

The system utilizes **Ridge Regression** (`sklearn.linear_model.Ridge`).

- **Why Ridge Regression?** This algorithm is chosen for its ability to handle multicollinearity (when independent variables are highly correlated) and to prevent overfitting through \$L2\$ regularization.
- **Model Serialization:** The trained model is saved as a binary file (`ridge_regression_model.pkl`) using the pickle library, allowing it to be reloaded for inference without retraining.

6. Application Interface (UI)

The user interface is built using **Streamlit**, a Python library for creating data apps.

User Inputs:

The application accepts the following parameters from the user:

1. **Experience:** A floating-point input with a step size of 0.1 years.
2. **Age:** An integer input constrained between 18 and 65 years.
3. **Education Level:** A numeric encoding system (0=High School, 1=Bachelor, 2=Master, 3=PhD).
4. **Job Level:** An integer input ranging from 1 to 5.

Prediction Logic:

Upon actuating the "Predict Salary" button, the system aggregates the inputs into a NumPy array, passes them through the loaded Ridge Regression model, and displays the estimated salary formatted to two decimal places.

7. Software Requirements

To replicate or run this project, the following Python libraries are required:

- `numpy` (Numerical computing)
- `pandas` (Data manipulation)
- `scikit-learn` (Machine learning algorithms)
- `streamlit` (Web application framework)
- `pickle` (Model serialization)